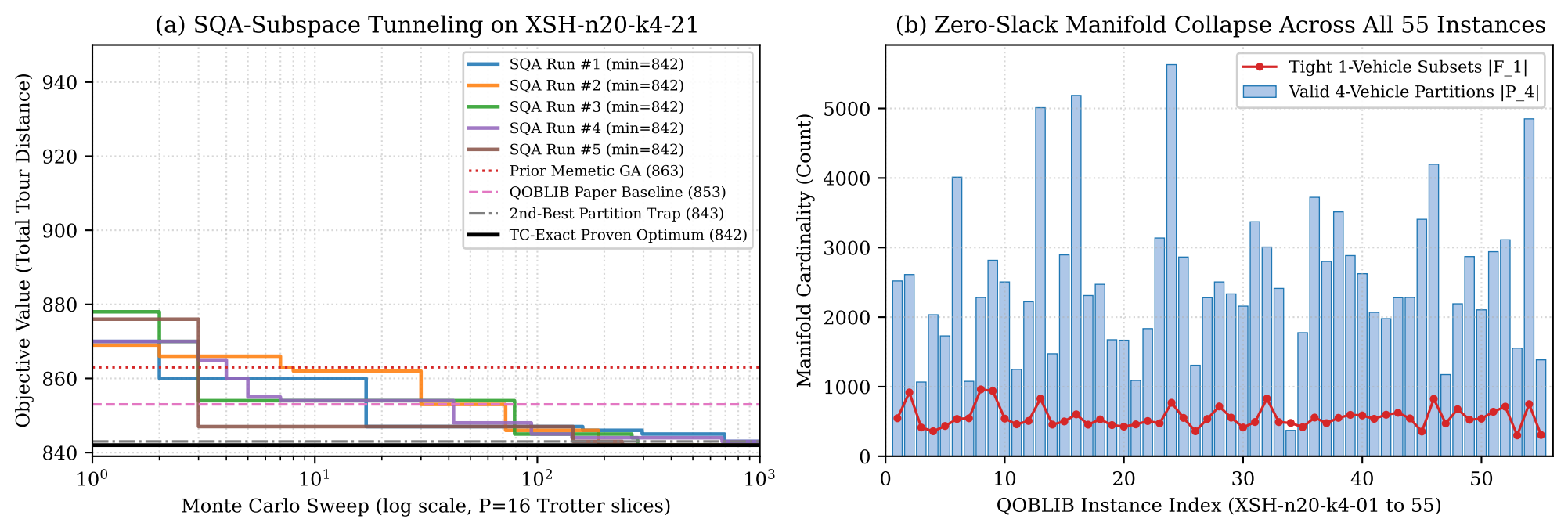
(a) SQA-Subspace Tunneling on XSH-n20-k4-21
SQA Run #1 (min=842)
SQA Run #2 (min=842)
SQA Run #3 (min=842)
SQA Run #4 (min=842)
SQA Run #5 (min=842)
Prior Memetic GA (863)
QOBLIB Paper Baseline (853)
2nd-Best Partition Trap (843)
TC-Exact Proven Optimum (842)
Objective Value (Total Tour Distance)
940
920
900
880
860
840
10^0
10^1
10^2
10^3
Monte Carlo Sweep (log scale, P=16 Trotter slices)
(b) Zero-Slack Manifold Collapse Across All 55 Instances
Tight 1-Vehicle Subsets |F_1|
Valid 4-Vehicle Partitions |P_4|
Manifold Cardinality (Count)
5000
4000
3000
2000
1000
0
0
10
20
30
40
50
QOBLIB Instance Index (XSH-n20-k4-01 to 55)